uc3m | Universidad Carlos III de Madrid

Master in Big Data

TECHNOLOGICAL FUNDAMENTALS IN THE BIG DATA WORLD

Lab 3. Web Services

INTRODUCTION

REST Web services are used to access data from one electronic device to another one using internet. This information can be in different formats such as JSON or XML.

The goal of this lab is to write Python programs that can retrieve information form a web portal using web services. More specifically, we are going to implement access using this technology in the EuroSTAT and Idescat websites using Python.

In order to obtain the different values from the indicators requested, which are going to be read in Python, you have to:

1.- Access the indicated web portal and look for the query builder.

2.- Build a query (or several, as you need) to extract the requested data.

3.- Include your query/ies in a Python program to extract and plot the data.

EXERCISE 1 - EUROSTAT

You have to get the balance data for the Consumer confidence indicator, that is seasonally adjusted, for the EU and work with them to print summarized bar plots.

More specifically, you have to print:

- The average EU consumer confidence for the last 5 years.
- The average EU consumer confidence for each month of 2024.
- The average Spanish consumer confidence for each month of 2024.

In order to obtain the different values from the consumer confidence indicator which are going to be read in Python, you have to:

1.- Access the EUROSTAT web page and look for the query builder.

2.- Build a query (or several, as you need) to extract the requested data.

Use the following parameters to setup the query builder:

- Database: Consumer surveys - monthly data (*ei_bsco_m)*
- Indicator: Consumer confidence indicator (*BS_CSMCI)*
- Adjusted: Seasonally adjusted data (*SA)*
- Unit: *Balance (BAL)*
- Time selection: Default
- Geo selection: Default
- Unit Code: Default
- Grouped indicators: NO
- Excluded dataset title: NO
- Excluded EU Aggregated: NO
- Number of Decimals: 1

Include your query/ies in a Python program to extract the data. Once the JSON's information has been read in Python, you can explore the data provided and build the data to be plotted.

EXERCISE 2 - IDESCAT

In this case, we want to get statistical information from the last 5 years of:

- Women born in Catalonia.

- Those with name María.
- The city of Catalonia where more Marías were born.

To build the query/ies, access to “Idescat.cat”, Servicios, API. Tablas: Onomástica.

Atributes:

- Operation: *nadons*
- Format: *json*
- Parameters:
  - id=40683
  - lang=en
- class: t

Create the Python program to extract, process and plot the information. You have to provide:

1. Total rank of women born in last 5 years
2. Total rank of Marias among females.
3. Frequency of the name.
4. Newborns named Maria/María per thousand.
5. Newborns named Maria/María per thousand among females.

LAB DELIVERY

Make the lab in groups of maximum 4.

- ☐ To deliver:
  - Authors.txt
  - One Python program per exercise

        - ws1.py
        - ws2.py
    - Report: max. 8 pages
- Deadline: **20 October 2025 – 23:30 hours**